

Unit 3, Lesson 6: Comparing a Quantity and Its Opposite in Real-World Context



Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.



Learning Targets

- ☐ I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context.
- ☐ I can use the structure of a horizontal and vertical number line to plot rational numbers from a real-world context.



3.6.1 Warm-Up: Ordering Elevations

Four images from locations across the United States are shown.

1. Write an integer that represents each elevation.

a.



Integer: _____

b.



Integer: _____

c.



Integer: _____

d.



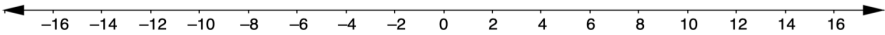
Integer: _____

2. Order the letter of the pictures from lowest elevation to highest.



3.6.2 Collaboration: A Quantity and Its Opposite

- 1.
- a. Plot the number 8 and its opposite on the number line.



- b. What do you notice about the distance of 8 and its opposite from zero?

2. Complete the table for each situation given.

	78.3 Feet Above Sea Level	32°C Below Zero	Withdrawal of \$27.68
Rational Number			
Opposite			
Meaning of Zero			



3. Omar is at the bookstore on Florida Avenue buying a book about training dogs. He also plans to visit Nolan's Pet Shop and Rover's Pet Shop to compare prices for a new leash for his dog. All three stores are on the same street. Nolan's Pet Shop is 7 blocks west of the bookstore. Rover's Pet Shop is 7 blocks east of the bookstore.

- a. Model the locations of the three stores in relation to each other on the number line.

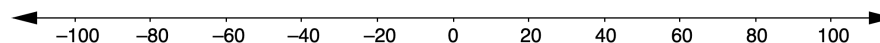
West ← ————— → East

- b. What does zero represent in this situation?
- c. Which pet store is farther away from the bookstore? Explain how you know.

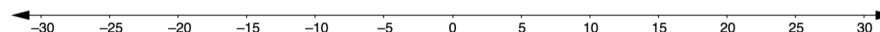


3.6.3 Try It: A Quantity and Its Opposite

1. On the number line, plot and label a deposit of \$64.87 and a withdrawal for the same amount from a bank account. What does 0 represent in this situation?



2. On the number line, locate and label 17°C below zero and 17°C above zero. What does 0 represent in this situation?



3. Carlos is at the ice cream shop. His house is 9.5 blocks north of the shop. The park is 9.5 blocks south of the ice cream shop.
- Plot and label all three places on the number line.
 - When at the ice cream shop, is Carlos closer to the park or his house?



4. Write the rational number that represents the description, the rational number's opposite, and the meaning of the opposite in context of the description.

Description	Rational Number	Opposite	Meaning of Opposite
An atom's positive charge of 7			
A deposit of \$35.84			
3,540 feet below sea level			
A rise of 41.6°C			
A loss of 13.3 pounds			

5. With a partner, choose one description and describe what 0 could represent in that situation.



3.6.4 Bring It Together: A Quantity and Its Opposite

Work with a partner to write answers to the following questions.

1. What is the relationship between any number and its opposite when plotted on a number line?
2. How can you use this relationship to locate the opposite of a given number on the number line?



3.6.5 Your Turn!

1. Janney notices that her room's temperature is 4°F below the recommended temperature, and Ivan's bedroom temperature is 4°F above the recommended temperature. Plot the values on the number line and explain what zero represents in this situation.



2. Finn wants to visit his friend Simon who lives 22.7 blocks west. He also wants to visit his grandparents who live 22.7 blocks east.

- a. Plot and label the locations of Simon and Finn’s grandparents on the number line.



- b. What does zero represent in this situation?

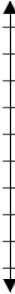
- c. Does Finn live closer to his grandparents or Simon?



3.6.6 Wrap-Up: Ticket Out the Door

Complete the following in the table.

- Write the rational number that represents the situation.
- Write the opposite of the rational number.
- Represent the rational number and its opposite on the number line with appropriate labels.
- Explain the meaning of zero in the situation.

	A Deposit of \$9.34	17°C Below Zero
Rational Number		
Opposite		
Number Line		
Meaning of Zero		

